

- 1 If $A \subseteq \Omega$, then $P(A) \geq 0$.
- 2 $P(\Omega) = 1$.
- 3 If $A_1, A_2, \dots$ are **mutually disjoint** i.e. $P(A_i \cap A_j) = 0 \quad \forall i \neq j$, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Properties of Probabilities

- $P(\emptyset) = 0$.
- $P(A') = 1 - P(A)$, where A' is the complement of A .
- If $A \subseteq B$, then $P(A) \leq P(B)$.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- For an increasing sequence of events $A_1 \subseteq A_2 \subseteq \dots \subseteq A_n \subseteq \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n \rightarrow \infty} P(A_n).$$

Definition (Conditional Probability)

Let A and B be two events with $P(B) \neq 0$. The conditional probability of A given B is defined to be

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Corollary (Multiplicative Rule)

$$P(A \cap B) = P(A | B)P(B)$$

General Case: For any event A_i such that $P(A_i) > 0$ for all $i = 1, \dots, n$

$$P\left(\bigcap_{i=1}^n A_i\right) = P\left(A_n \middle| \bigcap_{i=1}^{n-1} A_i\right) \cdots P(A_3 | A_2 \cap A_1) P(A_2 | A_1) P(A_1).$$

Proof: *By Induction.*

Inclusion-Exclusion Principle

For events A, B, C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) \\ - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

(General) Inclusion-Exclusion Principle

For any events $E_1, E_2, \dots, E_n$

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i) - \sum_{i < j} P(E_i \cap E_j) + \sum_{i < j < k} P(E_i \cap E_j \cap E_k) \\ + \dots + (-1)^{n-1} P\left(\bigcap_{i=1}^n E_i\right).$$

Law of Total Probability

Let $B_1, B_2, \dots, B_n$ be such that $\bigcup_{i=1}^n B_i = \Omega$ and $B_i \cap B_j = \emptyset$ with $P(B_i) > 0$ for all i , Then, for any event A :

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i)$$

Bayes' Rule

The conditional probability of B_j given A :

$$P(B_j | A) = \frac{P(A | B_j)P(B_j)}{\sum_{i=1}^n P(A | B_i)P(B_i)} = \frac{P(A | B_j)P(B_j)}{P(A)}$$